Singleton

Season: 1 · Creational · Category: GoF · Level: Beginner

Intent

- One shared instance with a global access point.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- One official passport office for the country.

When to Use

- True process-wide shared resource (metrics, feature flags).
- The design force named in Intent shows up more than once.

When Not to Use

- Request-scoped state; anything you need to mock easily.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.gof.creational.singleton;

import java.time.Instant;

/** Prefer injecting a Clock over a time singleton. */
public interface Clock {
    Instant now();

    record SystemClock() implements Clock {
        @Override public Instant now() { return Instant.now(); }
    }

    record FixedClock(Instant instant) implements Clock {
        @Override public Instant now() { return instant; }
    }
}
```

Modern Java 25 / Functional

- Inject Clock instead of System.currentTimeMillis(); use virtual threads safely with holder idiom.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/creational/singleton/
```

Pitfalls

- Enum singleton is preferred.
- Mutable singleton state is a concurrency hazard.
- Prefer DI at edges.

Interview Q&A

- Q: Enum vs double-checked locking? A: Enum is serialization/reflection-safe; DCL is easy to get wrong — prefer holder or enum.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- One shared instance with a global access point.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: One official passport office for the country. [0:40] Intent — State the design force in one sentence...